



Pre-Hackathon

Coding Challenge

Smart Invoice Processing System

Welcome, developers, to the hackathon! This document outlines the official challenge. We're looking for teams that can think critically about real-world problems and build practical, elegant solutions. Your mission is to tackle the challenge described below.

1. Background

Businesses and individuals alike often struggle with managing various types of invoices—from vendor bills and personal expenses to contractor payments and other financial documents. The traditional process involves manually extracting data from PDF invoices, entering information into spreadsheets, preparing payment files for banking software, and tracking which invoices have been paid. This manual workflow is time-consuming, prone to error, and becomes increasingly unmanageable as the volume of invoices grows.

2. The Challenge

Your task is to design and build a web-based system that helps users automate their invoice processing and payment preparation workflow. Think of it as replacing the tedious manual work of copying invoice details into Excel sheets and preparing payment files for various types of payments—be it for business expenses, personal bills, or contractor payments.

3. Core Requirements

Your task is to design and build a web-based system that helps users automate their invoice processing and payment preparation workflow. Think of it as replacing the tedious manual work of copying invoice details into Excel sheets and preparing payment files for various types of payments—be it for business expenses, personal bills, or contractor payments.

3.1. Invoice Data Extraction

- Users must be able to upload invoice documents (**PDFs** or **images**)



- The system should automatically extract key information, including:
 - Invoice amount and currency
 - Payee details and payment information (vendor, landlord, service provider, etc.)
 - Invoice date and due date
 - Tax / VAT information where applicable
- Consider how to handle edge cases, such as poor-quality scans or multiple invoices within a single file.

3.2. Invoice Management

- Store and organize all uploaded invoices in a structured manner.
- Allow users to review and, if necessary, correct the extracted data.
- Track the status of each invoice (e.g., Pending, Approved, Paid).
- Support invoices from different sources (business vendors, personal bills, contractors)
- and for different entities or accounts.

3.3. Payment Preparation

- Enable users to export payment data as **CSV files** (UTF-8 encoded), ready for import into banking software.
- The export must include core payment fields: payee name, amount, account number, reference, and due date.
- Generated files should be compatible with common online banking CSV formats.

4. Constraints & Considerations

- The solution must be **web-based** and accessible from any device.
- Focus on **user experience**. Your users may not be tech-savvy, so the interface should be intuitive and simple.
- The system should be robust enough to handle common invoice formats from various global sources

5. Recommended Technology Stack

- **Backend:** Firebase (Firestore, Cloud Storage, Authentication, Cloud Functions)
- **AI/ML:** Google Gemini API
- **Frontend:** Your choice of modern web framework (e.g., React, Vue, Svelte, Angular)

6. Bonus Features (Optional)

- **Multi-language support** for invoices in different languages.
- A fully **responsive, mobile-friendly interface** for on-the-go management.
- **Bulk operations** (e.g., bulk approval, bulk export) to improve user efficiency.



7. Evaluation Criteria

Your submission will be evaluated based on the following criteria. We are interested in seeing how you think about the problem and translate those thoughts into a functional and user-friendly product.

Criteria	Weighting
Problem Understanding	25%
Technical Implementation	25%
User Experience (UX)	25%
Innovation	15%
Completeness	10%

Note: This is not just about building features. We're looking for a deep understanding of the user's pain points and a practical solution that provides real value.

8. Deliverables

Please submit the following items via email to hi@morromakers.com. The deadline for this challenge is the **07th of August 2025**.

1. **A working web application**, either deployed to a public URL or with clear, concise setup instructions in a README.md file.
 2. **Brief documentation** explaining your architectural approach, key features, and any trade-offs you made.
 3. **A short demo video (3-5 minutes)** showcasing the system in action, from uploading an invoice to exporting a payment file.
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9. Hints for Success

- Think deeply about what makes invoice processing painful today.
- Focus on the core functionality that provides the most immediate value to a user.
- Consider the different types of invoices people deal with in their daily lives (e.g., a freelancer's invoice vs. a utility bill).
- Leverage the unique capabilities of modern AI to go beyond simple text extraction.

Good luck! We're excited to see what you build and hopefully meeting you soon in person in Rio de Janeiro.